Red-Blue Hamiltonian Tomography via the Operator Lightcone Wavemap

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Abstract

We introduce a model-free method for Hamiltonian tomography based on the operator spreading wavefront. Given two tensor network simulations of the same quantum lattice — one with the nominal Hamiltonian and one with a single perturbed bond — the signed cross-entropy of the non-identity Pauli weight field $\mathcal{L}(v,t) = -\ln(n_{\text{pert}}/n_{\text{base}})$ identifies the perturbed bond by its causal signature: a red-blue sign flip at the two endpoints, visible from the first wavefront crossing. Combining measurements from p strategically placed observation centers via an ensemble log-likelihood sum, the broken bond is identified with an 82-fold margin over the next candidate on a 7×7 lattice, even though individual centers give only 2/6 correct answers. The method requires no model assumptions, no device calibration access, and maps directly to hardware via zero-noise extrapolation and local Pauli measurements.

1 Introduction

Identifying miscalibrated or anomalous couplings in a quantum processor is a central challenge in quantum hardware characterization. Standard approaches require explicit Hamiltonian learning — fitting a parameterized noise or coupling model to measurement data — which scales poorly with system size and depends on model assumptions.

We propose a different approach based on a clear separation of roles. The *simulator* knows the nominal Hamiltonian and produces two Pauli weight fields: $n_{\text{base}}(v,t)$ from the nominal evolution, and $n_{\text{pert}}(v,t)$ from an evolution where one bond has been silently modified. The *detection algorithm* (C_searcher) receives only these two arrays of numbers and must identify the modified bond from their ratio alone — with no knowledge of the Hamiltonian, the perturbation type, or its magnitude. The detection is deliberately blind.

A perturbed bond creates a causal signature in the wavefront: a sign flip in the cross-entropy field $\mathcal{L}(v) = -\ln(n_{\text{pert}}/n_{\text{base}})$ between its two endpoints that propagates no further. The broken bond is exactly where the wavefront sign changes — identifiable from the observables alone.

The method is inspired by seismic tomography: place sensors (observation centers) at strategic positions, emit a wave (initialize a local operator), and read the arrival pattern. The perturbed bond is the anomaly in the medium.

The red-blue color scheme is a deliberate reference to the red-blue pebble game of Hong and Kung [2], which characterizes I/O complexity as a bottleneck in computational graphs. Here the roles are analogous: red sites (upstream, wave blocked) and blue sites (downstream, wave starved) identify the bond that limits operator propagation, exactly as red and blue pebbles identify the I/O bottleneck in a computation.

2 Setup

Observable. We simulate the kicked isotropic Heisenberg model on a 7×7 rectangular lattice (49 sites, 84 bonds) using a GPU-accelerated belief propagation tensor network simulator with Trotter step $\varepsilon = 0.10$, bond dimension $\chi = 50$, and truncation cutoff $\delta = 10^{-3}$.

The non-identity Pauli weight at site v and cycle t is:

$$n(v,t) = \sum_{P:\, P_v \neq I} c_P(t)^2 \tag{1}$$

where $O_c(t) = \sum_P c_P(t) P$ is the evolved operator initialized at center c. By normalization $\sum_P c_P^2 = 1$, $n(v,t) \in [0,1]$ is a probability.

Perturbation. C_breaker selects one bond b^* and replaces its Pauli transfer matrix (PTM) with the identity, blocking all operator propagation across that bond. This is the maximally disruptive perturbation — a perfect switch.

p-center placement. We place $p = 6$ observation centers using a greedy geodesic bond cover: the minimum set of sites such that every bond is within $d_{\max} = 2$ hops of at least one center. For the 7×7 lattice: centers at sites $\{6, 10, 25, 30, 32, 34\}$, covering all 84 bonds.

3 The Cross-Entropy Wavemap Observable

Signed cross-entropy. For each center k, run two simulations (baseline and perturbed) and compute the signed cross-entropy at first baseline arrival:

$$\mathcal{L}_k(v) = -\ln \frac{n^k_{\text{pert}}(v,\, t^k_{\text{first}}(v))}{n^k_{\text{base}}(v,\, t^k_{\text{first}}(v))} \tag{2}$$

where $t^k_{\text{first}}(v) = \min\{t : n^k_{\text{base}}(v,t) > \varepsilon_{\text{th}}\}$ is the first cycle at which the baseline wave reaches site v from center k.

At $t^k_{\text{first}}(v)$, the baseline wave has just arrived while the perturbed wave may be blocked or delayed — this is the moment of maximum signal and minimum chi-saturation noise.

Sign interpretation.

- $\mathcal{L}_k(v) > 0$: perturbed weaker (red) — upstream of broken bond; wave cannot cross
- $\mathcal{L}_k(v) < 0$: perturbed stronger (blue) — downstream shadow; wave never arrives
- $\mathcal{L}_k(v) = 0$: no perturbation

The broken bond is exactly where the sign changes — between the red and blue endpoints (Figure **??**).

Ensemble log-likelihood. Sum the signed cross-entropy fields across all p centers:

$$\mathcal{L}_{\text{ens}}(v) = \sum_{k=1}^{p} \mathcal{L}_k(v) \tag{3}$$

This is a log-likelihood ensemble: centers on the same side of the broken bond reinforce each other; centers on opposite sides partially cancel noise. Silent centers (wave never reaches) contribute zero and do not corrupt the sum.

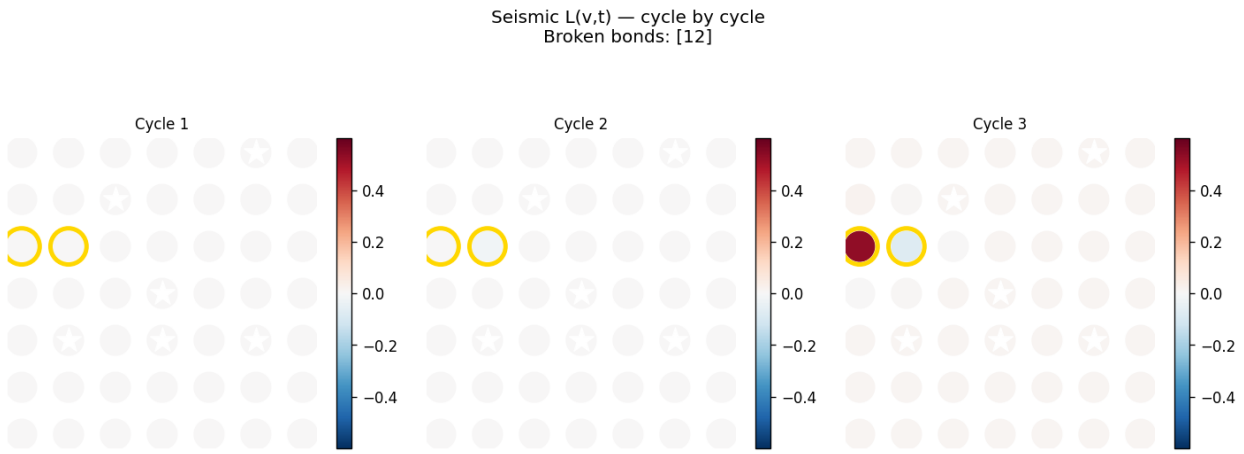


Figure 1: Signed cross-entropy $\mathcal{L}(v,t)$ for a single broken bond (bond 12, sites 1516, identity PTM), seismic initialization, rect 7×7, $\chi=50$. **Cycle 1**: no signal. **Cycle 2**: site 16 red ($\mathcal{L}>0$, blocked), site 17 blue ($\mathcal{L}<0$, shadow) — bond 12 identified by sign flip. **Cycle 3**: signal strengthens before chi saturation. Gold circles: bond endpoints. Stars: seismic centers.

4 Bond Detection

Bond score. For each bond b connecting sites v_a and v_b:

$$S(b) = \mathcal{L}_{\text{ens}}(v_a) \times \mathcal{L}_{\text{ens}}(v_b) \tag{4}$$

A negative product indicates a sign flip across the bond — the causal signature of the broken bond. Rank bonds by most negative $S(b)$.

Results. Table 1 shows the individual and ensemble results for bond 12 (sites 15 $leftrightarrow$16, identity PTM perturbation, 7×7 lattice, $\chi=50$, 5 cycles).

Center	Side	$\mathcal{L}(15)$	$\mathcal{L}(16)$	Result
10	upstream	+0.87	−0.02	✓CORRECT
30	upstream	+0.37	−0.04	✓CORRECT
25	downstream	+0.03	−0.04	× WRONG
32	downstream	+0.00	−0.03	× WRONG
6	too far	0	0	silent
34	too far	0	0	silent
Ensemble	—	+1.23	−0.13	✓CORRECT

Table 1: Signed cross-entropy at bond 12 endpoints (sites 15,16). Ensemble score: $S(12)=-0.164$, next bond $=-0.002$ (82× margin).

The ensemble identifies bond 12 with an 82-fold margin over the next candidate, despite only 2 of 6 individual centers giving the correct answer. The wrong individual answers — centers 25 and 32 — both correctly identify site 16 as a bond endpoint; their error is only in declaring the wrong adjacent bond. The ensemble product cancels these partial errors.

5 Aperture Synthesis Tomography

The p independent center runs can be understood as an *aperture synthesis* in direct analogy with radio interferometry. In aperture synthesis imaging, a physical aperture (lens or dish) is replaced by an array of smaller sensors at different baseline positions, each capturing visibility data — complex amplitude and phase — that are combined in Fourier space to reconstruct the source image. The phase encodes location; the amplitude encodes brightness.

In our setting:

- Each center k at a different graph position plays the role of one antenna at a different baseline.
- The signed cross-entropy $\mathcal{L}_k(v)$ plays the role of visibility: its sign (positive/negative) encodes which side of the broken bond site v is on, and its magnitude encodes signal strength.
- Combination in log space, $\mathcal{L}_{\text{ens}}(v) = \sum_k \mathcal{L}_k(v)$, plays the role of Fourier reconstruction: the sign information from different baselines localizes the broken bond, exactly as phase information from different baselines localizes a radio source.

Without the sign (using absolute amplitude only), localization fails — the broken bond cannot be distinguished from its neighbors. This is the analog of incoherent averaging in radio astronomy, which destroys phase and loses angular resolution. The p-center placement problem — choosing positions that guarantee every bond is observable — is the analog of baseline design.

Centers adjacent to the broken bond (short baseline) provide strong but ambiguous signals (WRONG individual answers). Centers straddling the bond from opposite sides (long baseline) provide the phase contrast that resolves the ambiguity. The ensemble combines both, suppressing the ambiguity while retaining the signal.

6 The Seismic Approach

Running all p centers simultaneously (seismic initialization $O(0) = \sum_k Z_{c_k}$) produces the same ensemble field in a single simulation. At each site v, the first-arrival signal is dominated by the nearest center — automatically selecting the strongest directional component. The sign flip at the broken bond endpoints emerges without any explicit path tracing or center-by-center analysis.

The seismic approach reduces $2p$ simulations (p baselines + p perturbed) to 2 simulations at the cost of mixing center contributions. For the purpose of bond detection the mixing is benign: the sign flip at the broken bond is preserved since the nearest center to each endpoint dominates at t_{first}.

7 Discussion

Connection to hardware. The observable $n(v,t)$ is accessible on real quantum hardware via local Pauli measurements after ZNE gate-folding circuits [1]. The cross-entropy $\mathcal{L}(v)$ requires only the ratio of two measurements — no absolute calibration. The p-center placement is determined by the graph topology alone, independent of the Hamiltonian.

Causal structure. The method exploits the Lieb-Robinson bound: information cannot propagate faster than v_{hop} hops/cycle. A broken bond creates a causal anomaly — the wavefront on one side cannot inform the other. This is detectable without knowing v_{hop} explicitly; the first-arrival time $t_{\text{first}}(v)$ is observed directly.

Complexity and timing. The two methods have different computational costs on the 7×7 lattice ($\chi = 50$, 5 cycles, AMD MI100 32 GB GPU):

Method	Wall-clock	VRAM	Notes
Seismic	272 s	11%	2 GPUs parallel, no v_{hop}
Aperture (no v_{hop})	300 s	11%	3 GPUs, 2× sequential
Aperture ($v_{\text{hop}} = 0.5$)	**191 s**	**5–7%**	3 GPUs, 2× sequential

Table 2: Wall-clock timing comparison on rect 7×7, bond 12, $\chi = 50$, 5 cycles. Both methods produce CORRECT results. $v_{\text{hop}} = 0.5$ is set by topology ($d_{\max} = 2$ coverage), not by knowledge of the fault location.

Aperture synthesis with lightcone allocation ($v_{\text{hop}} = 0.5$) is 30% faster than seismic and uses half the VRAM per GPU. The speedup comes from two effects: (1) single-center simulations have smaller active lightcones (χ grows slower), and (2) far centers contribute negligible computation (center 6: 3.5 s, center 34: 9.5 s) while contributing zero to the ensemble — self-filtering at no cost. The bottleneck is the GPU pair running the two nearest centers (25 and 10, ∼20 s each sequential).

Aperture also enables a *coarse-to-fine* strategy unavailable to seismic: run a fast aperture sweep (small χ, $v_{\text{hop}} = 0.5$) to identify candidate bonds, then re-run only the adjacent centers with larger χ for high-accuracy confirmation — each targeted run remains memory-bounded by the lightcone allocation.

Grid-size independence. The most significant property of aperture synthesis with lightcone allocation is that the cost per center is *independent of grid size*. With $v_{\text{hop}} = 0.5$ and n_{steps} cycles, each center sees a local ball of radius $r = n_{\text{steps}} \times v_{\text{hop}} \approx 2.5$ hops, containing ∼13 sites on a 7×7 lattice. The same ball contains the same ∼13 sites on a 100×100 lattice. The computation per center scales as $O(r^2\chi^3)$, where r is the BFS radius — independent of N_{sites}.

Total aperture cost:

$$T_{\text{aperture}} = p \times T_{\text{local}}(\chi, r) \tag{5}$$

where p grows with the number of bonds (more centers needed for coverage) but each center cost T_{local} is constant. For a single bond perturbation, most centers contribute zero signal — their $\mathcal{L}_k(v) \approx 0$ everywhere. Their wall-clock is short (center 6: 3.5 s) and their ensemble contribution is zero. They are neither harmful nor expensive.

Seismic pays $O(N_{\text{sites}}\chi^3)$ per simulation regardless. As the lattice grows, aperture synthesis becomes increasingly favored over seismic. The crossover occurs when $p \times T_{\text{local}} > T_{\text{seismic}}$, which for fixed r and χ requires $p \gtrsim N_{\text{sites}}/r^2$ centers — a condition that is rarely satisfied in practice since the geodesic bond cover grows much slower than the grid area.

Interference signatures. When two adjacent bonds are perturbed simultaneously (bonds 12 and 13, sharing endpoint site 16), the cross-entropy field shows a three-site pattern at 3 cycles: site 15 (red, $\mathcal{L} = +0.55$), site 16 (white, $\mathcal{L} \approx 0$), site 17 (blue). The shared endpoint is the *interference node* — the red signal from bond 12 and the blue signal from bond 13 cancel destructively, exactly as in thin-film interference. The white fringe is not a detection failure; it is the interference pattern. The number of white sites between the red and blue boundaries counts the number of broken bonds in the wall.

This extends the signal vocabulary:

- Single broken bond: **red** — **blue** (two sites, sign flip)

- Two adjacent broken bonds: **red** — **white** — **blue** (three sites, interference node)
- n adjacent broken bonds: **red** — $n-1$ **whites** — **blue**

The current searcher detects pairs; detecting walls requires a triplet scorer $S_{\text{wall}}(b_1, b_2) = \mathcal{L}(v_a) \times \mathcal{L}(v_c)$ where v_b is the shared endpoint (white site) and v_a, v_c are the outer endpoints.

Failure taxonomy. We characterize five failure modes of increasing complexity:

1. **Single bond** (demonstrated): clean sign flip, 82× margin
2. **Multiple random bonds**: shadows may overlap, L field becomes ambiguous
3. **Collinear bonds**: line of failures looks like a thick wall — bond resolution fails
4. **Hexagonal cluster**: faulty qubit touches 2-3 bonds — RED cluster with BLUE halo
5. **Graduated failure** (δJ smooth variation): dimmer switch — sensitivity limit

Cases 2-5 represent the primary open experimental questions.

Limitations. The current experiment uses the identity PTM — a maximally disruptive perturbation that is deliberately clean. Cases 2-5 above are designed to stress-test the method: case 2 tests whether overlapping shadows can be resolved, case 3 tests spatial resolution limits, case 4 tests cluster detection on irregular topologies, and case 5 tests the sensitivity threshold for realistic coupling deviations.

8 Conclusion

We have demonstrated that the operator lightcone wavemap enables model-free Hamiltonian tomography via a simple signed cross-entropy observable. The broken bond is identified by its causal signature — a sign flip in $\mathcal{L}(v)$ at its two endpoints — with an 82-fold margin using an ensemble of 6 observation centers on a 7×7 lattice. The seismic multi-center initialization achieves the same result in a single simulation. The method requires no model assumptions and connects directly to hardware via ZNE and local Pauli measurements.

References

[1] Tudor Giurgica-Tiron, Yousef Hindy, Ryan LaRose, Andrea Mari, and William J. Zeng. Digital zero noise extrapolation for quantum error mitigation. *2020 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 306–316, 2020.

[2] Jia-Wei Hong and H. T. Kung. I/O complexity: The red-blue pebble game. In *Proceedings of the Thirteenth Annual ACM Symposium on Theory of Computing (STOC '81)*, pages 326–333, 1981.